

Occupancy API

Occupancy API can be used to create occupancy profiles for either the whole building or per floor. This API is under development and all parts of it are subject to change.

Common parameters

API endpoints have common request parameters:

- Headers
 - "Authorization" - The API key that allows accessing the API for a specific site (usually, one building).
- Query parameters
 - "start-date" - ISO date string specifying the start date for profile calculation period. Mandatory.
 - "end-date" - ISO date string specifying the end date for profile calculation. Mandatory.
 - "weekends" - if present and set to value "0", then weekends will be stripped from the results. Optional.
 - "normalize" - if present and set to value "0", then result normalization to range 0-10 will not be done. Optional.
 - "resolution" - Selects the time resolution of the occupancy profile. Allowed values are 60, 30, and 15 (minutes). Default is 60 minutes if the parameter is not present. Optional.
 - "remove_residual_occupancy" - if present and set to values "1", then the possible evening residual occupancy is removed by balancing the exits to match the entries over the day.

Building profile

Endpoint: api/occupancy/building

Response:

List of JSON objects like:

```
[
  {
    "month": 1,
    "weekday": 1,
    "site_connection_ok": true,
    "occupancy": [1, 4, 8, ...]
  },
  {
    "month": 1,
    "weekday": 2,
    ...
  },
  ...
]
```

Where:

- list contains one element per weekday per month
- "month" - number in range 1-12, where 1 = January, 2 = February and so on
- "weekday" - number in range 1-7 where 1 = Monday, 2 = Tuesday and so on
- "site_connection_ok" - boolean or null, false indicating there has been a connection issue during the requested time period. Connection data is available only for past 30 days, historical data will have null as value.
- "occupancy" - number in range 0 - 10 which indicates normalized busyness for every time period (depending on the selected resolution) during the day. Value 10 means most crowded. First element starts at 00:00:00 and last element ends at 23:59:59.

Floors profile

Endpoint: api/occupancy/floors

Response:

List of JSON objects like:

```
[
  {
    "month": 1,
    "weekday": 1,
    "site_connection_ok": true,
    "floor": "1",
    "floor_id": 3,
    "occupancy": [1, 4, 8, ...]
  },
  {
    "month": 1,
    "weekday": 2,
    "floor": "2",
    "floor_id": 4,
    ...
  },
  ...
]
```

Where:

- list contains one element per weekday per floor per month
- "month" - number in range 1-12, where 1 = January, 2 = February and so on
- "weekday" - number in range 1-7 where 1 = Monday, 2 = Tuesday and so on
- "site_connection_ok" - boolean or null, false indicating there has been a connection issue during the requested time period. Connection data is available only for past 30 days, historical data will have null as value.
- "floor" - floor the profile is for
- "floor_id" - abstract floor ID for the floor, lower value means lower floor
- "occupancy" - number in range 0 - 10 which indicates normalized busyness for every time period (depending on selected resolution) during the day. Value 10 means most crowded. First element starts at 00:00:00 and last element ends at 23:59:59.

Examples

The example gets floors profile for one month from the site the API key has access to.

```
export AUTH='my-token'
export BASE_URL='https://api.production.siteflow.kone.com'

curl "$BASE_URL/api/occupancy/floors?start-date=2025-03-01&end-date=2025-04-01" -H "Authorization: $AUTH" --silent | jq .
[
  {
    "month": 3,
    "weekday": 1,
    "site_connection_ok": true,
    "floor": "1",
    "floor_id": 3,
    "occupancy": [
      1,
      4,
      8,
      ...
    ]
  },
  {
    ...
  },
  ... weekdays 1-7 ...
]
```

Get one month building profile from imaginary site with weekends dropped from the results and without normalization:

```
export AUTH='my-token'
export BASE_URL='https://api.production.siteflow.kone.com'

curl "$BASE_URL/api/occupancy/building?start-date=2025-03-01&end-date=2025-04-01&weekends=0&normalize=0" -H "Authorization: $AUTH" --silent | jq .
[
  {
    "month": 3,
    "weekday": 1,
    "site_connection_ok": true,
    "occupancy": [
      3,
      123,
      249,
      ...
    ]
  },
  {
    ...
  },
  ... weekdays 1-5 ...
]
```

Get the data for one day without normalization at 15 min resolution.

```
export AUTH='my-token'
export BASE_URL='https://api.production.siteflow.kone.com'

curl "$BASE_URL/api/occupancy/floors?start-date=2025-04-01&end-date=2025-04-02&normalize=0&resolution=15" -H "Authorization: $AUTH" --silent | jq .
[
  {
    "month": 4,
    "weekday": 2,
    "site_connection_ok": true,
    "floor": "1",
    "floor_id": 3,
    "occupancy": [
      0,
      4,
      8,
      ...
    ]
  },
]
```

```
{  
  ...  
},  
]
```